Microfluidic Chip for Biosensing Yeast Cells

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ABSTRACT

Long-term monitoring of xenoestrogens, a human endocrine disruptor that cause serious health implications is needed. In this work, we design a microfluidic device capable of sustaining yeast populations over an extended period to monitor these disruptors.

CCS CONCEPTS

• **Hardware → Bio-embedded electronics**.

KEYWORDS

Molecular communication, Optogenetics, Microfluidic network,

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1 INTRODUCTION

Xenoestrogens are human endocrine disruptors introduced into rivers and lakes by municipal wastewater treatment plants. These disruptors cause serious health implications in humans and animals, such as infertility. Monitoring exposure is critical to managing aquatic ecosystems and human health. While existing procedures can determine concentrations within a short period of time, long-term monitoring can provide helpful insight into the effect of disruptors. The yeast species *Saccharomyces cerevisiae* detects estrogenic compounds and can be used to monitor the exposure of endocrine disruptors [1]. Microfluidic chips can be used to grow and monitor yeast populations (Figure 1). They are capable of housing populations using a perfusion system for feeding and waste removal. To test the results, estrogenic compounds are fed into the chip and the yeast biosensors will detect the presence of estrogenic compounds, producing fluorescence in response to estrogenic compounds that can be monitored under fluorescence microscopes.

Current autonomous cell culture methods are large scale, usually with around 50 ml of yeast and containing media. This setup is not portable, and cannot be placed under a microscope for inspection. The current setups are also very expensive, costing around $1000

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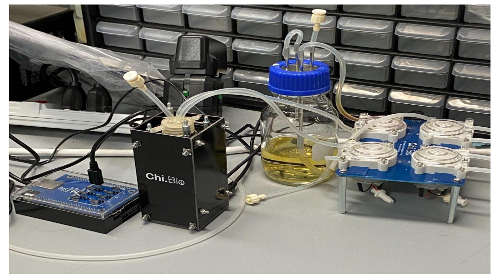

Figure 1: Macroscopic biosensing yeast cell culture setup - Tabletop setups requires human input to regulate media tank

for the base price, and more with peripherals. Lastly, current models within that price range are not completely autonomous, requiring human input to regulate the media tank (see Figure 1). **In this work, we design a microfluidic device capable of sustaining yeast populations over an extended period.** The device captures suspended yeast cells in wells connected to microfluidic channels providing cell media and test samples (Figures 2 3, 4). By flowing excess yeast cells out of the chip and continuously supplying new media into the chip, yeast cells can be maintained long-term for sensing of endocrine disruptors. We study three chip geometries to identify the best fit based on metrics such as cell distribution, isolation, capacity, ease of fabrication among others.

The microfluidic chip design in this work is based on the following four major requirements for long term sensing.

(1) Physical dimension : Maintain a microfluidic flow to supply media, remove excess cells, and expose cells to endocrine disruptors. (2) Chip material : The material used for the device must be biocompatible and be able to be engraved with micro-scale channels as well as be resistant to absorption of media, EDCs, and waste products. The material should not produce excessive background fluorescence, which would interfere with accurate detection. (3) Chassis material : The chassis must be made out of a water-resistant material that can be cheaply and easily fabricated. (4) Yeast density : The yeast density should be on average 3×107 cells/mL, with a maximum of 25% variation over the timescale of the experiment. The cell density can be measured via optical density (OD) at 450 nm, which is the excitation wavelength for the biosensing yeast cells, as well as with the use of trypan blue staining and a hemocytometer. We compare three designs for chip fabrication. The material considered is polydimethylsiloxane (PDMS), as it is cost efficient, flexible, durable, and bio-compatible [2].

2 DESIGN PRINCIPLES

All our designs assume the model of a constant stream of nutritional media to encourage a high rate of cell growth. This growth is offset by the fluid flow of the cell media which removes cells from the well. For a continuous culture of cells in a specified chamber, the

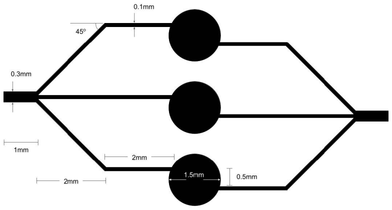

Figure 2: Triple Junction

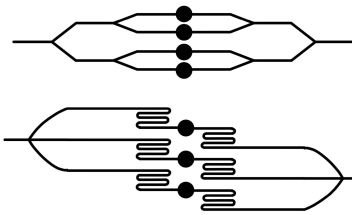

Figure 3: Coupled Y-Junction

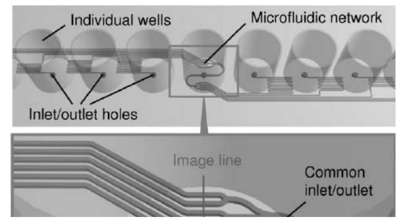

Figure 4: Serpentine Channel

accumulation of cells follows the following mass balance equation:

$$Accumulation = input - output + Generation - Consumption \quad (1)$$

In the device's steady-state, there will be no cells added and a constant stream of cell output due to flow. No cell death is expected due to the amount of cell media constantly being added. Thus this equation can be rewritten as: $\frac{dX}{dt} = \mu X - DX$ where, μ is the growth rate of cells, X the concentration of cells, and D being the dilution rate [3]. In order to maintain a constant population, no accumulation is expected, and $\frac{dX}{dt}$ =0. The relationship shown by this requires that the specific growth rate of the cells is equal to the dilution rate of the device, $\mu = D$. The dilution rate can be calculated from dividing the flow rate by the volume of the culture. The volume of the cell-housing wells can be described by the surface area of the well (either 1.5mm or 1.0mm in diameter) multiplied by the height of the well (100 μm). The flow rate can be controlled based on the feeding system and is equal to the fluid velocity times the cross-sectional area of the tubing. The diameter of the inlet is 0.8mm, this circular area can be assumed to be the cross-sectional area. The total volume of the wells is four times the volume of a single well, and all four wells are fed by one stream of flow. The specific growth rate is described by the Monod equation:

$$\mu = \mu_{max} \frac{[S]}{K_S + [S]}, \text{where [S] is the glucose concentration.} \quad (2)$$

Using the above design principles, we design three junctions viz., Triple Junction, Coupled Y-junction, and Serpentine Channel.

Triple Junction. The Triple Junction, illustrated in Figure 2 includes three streams of flow that originate from one primary channel. One end is used as the input and the other as the output. The cells would remain suspended in the center wells and the offset of the channels is designed to encourage cells to stay within the well.

Coupled Y-Junction. This design uses a nested cascade of Y-junctions stemming from one main channel to four wells. It only uses junctions that split symmetrically in two, which is a commonly used junction in microfluidics that has been proven to split flow equally. This design can only accommodate a number of wells that is a power of two, therefore, it has four wells.

Serpentine Channels. The Serpentine Channels design maintains three wells and utilizes a design element used by Choi and Cunningham [4] to equalize channel lengths in a microfluidic chip. This design makes use of curved, S-shaped pathways with varying number of curves to produce equal lengths in channels that are otherwise differing in length. Serpentine channels in other microfluidics can have the effect of immobilizing molecules flowing within the curves and acting as a flow resistor to prevent turbulence of flow. The center channel will have more swerving in its channels than the side channels as the center channel is the shortest.

3 PRELIMINARY RESULTS

We analyze the above three designs using the following 7 parameters with the weights in braces adding up to 100.

(1) Cell Distribution (25): A functional design must meet the requirement of consistent cell concentrations. (2) Isolation (20): Each well should maintain yeast cultures and not have any cross-contamination. (3) Input and Output (20): How well it can input media, output excess cells, flow velocity, and laminar flow. (4) Capacity (15): The more wells, the more cell colonies can be cultured. It may be desirable to observe multiple samples so as to obtain more information. (5) Ease of Fabrication (10): It refers specifically to the ease of creating the computer model of the design, which outlines the proper geometry that will then be made into a mold for fabrication. (6) Cost(10): Low cost will result in wide deployments.

Based on our score for each design, **we calculate triple junction to be 66, couple-y to be 74 and serpentine to be 70.** While both the coupled Y-junction and serpentine channel designs aim to equalize the path lengths for the channels. Y-channels are known to be reliable for splitting flow, so the Coupled Y-junction design is scored the highest. This design is expected to give the most reliable flow splitting for the requirement of consistency between each well and will be relatively easy to create the correct geometry for. We have included in the design an alternative with straight channels and an alternative with channels offset to the side of the wells. It is unknown which design can best trap cells into the well, and testing may be done in the future to confirm which design is most fitting.

4 CONCLUSIONS AND FUTURE WORK

We designed a microfluidic chip to feed and maintain colonies of biosensing yeast cells that secrete endocrine disrupting chemicals, including managing nutrient flow, population density, and waste removal. While preliminary results were obtained using COMSOL, future physical testing with the final PDMS chip product is needed to determine if our design is viable for the purpose of consistent cell culture. A detailed report to recreate our chip and further results from COMSOL resulting is available in our technical report [5].

REFERENCES

[1] Bayram Y, Hakan T, Suleyman S, and Fahrettin K. Endocrine disrupting chemicals: exposure, effects on human health, mechanism of action, models for testing and strategies for prevention. *Reviews in endocrine and metabolic disorders*, 2020.
[2] Rajendrani Mukhopadhyay. When pdms isn't the best, 2007.
[3] Ghasem Najafpour. *Biochemical engineering and biotechnology*. Elsevier, 2015.
[4] Charles J Choi and Brian T Cunningham. A 96-well microplate incorporating a replica molded microfluidic network integrated with photonic crystal biosensors for high throughput kinetic biomolecular interaction analysis. *Lab on a Chip*, 2007.
[5] Technical report. https://bmedesign.engr.wisc.edu/projects/s22/biosense_microfluidic.